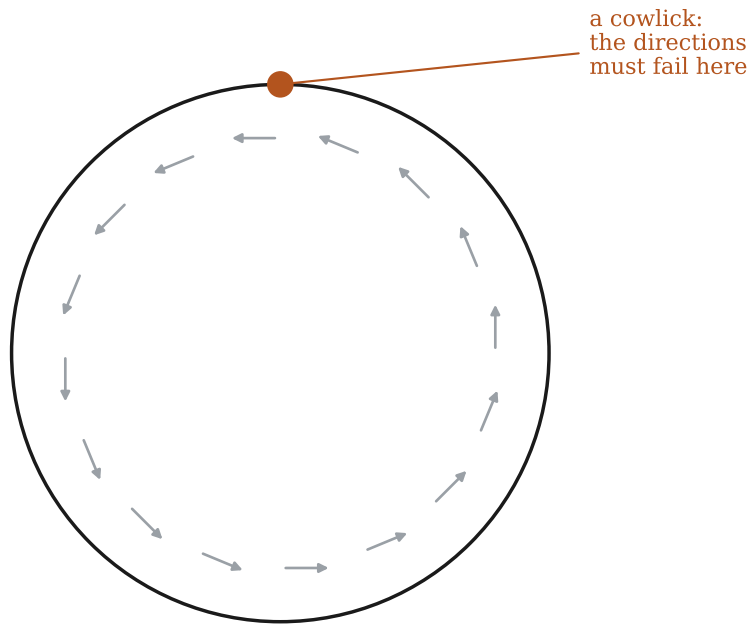
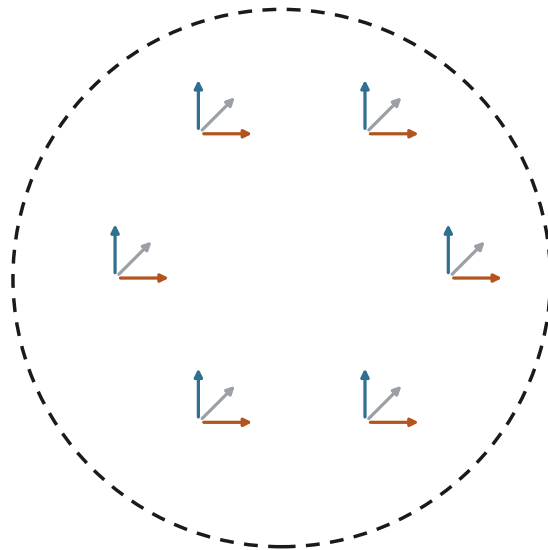


the 2-sphere can't be combed



no smooth set of directions covers all of it —
one point always fails

the 3-sphere (and the 7-sphere) can



a full frame of directions sits smoothly at every point —
three on the 3-sphere, seven on the 7-sphere